

72 The table shows the concentration of potassium ions in several different locations.

<i>location</i>	<i>concentration of potassium ions</i>
bacterial cell cytoplasm	30 mmol dm^{-3}
mammalian blood plasma	$4000 \text{ } \mu\text{mol dm}^{-3}$
mammalian heart cell cytoplasm	$1.0 \times 10^2 \text{ mmol dm}^{-3}$
sea water	$3.0 \times 10^4 \text{ } \mu\text{mol dm}^{-3}$
yeast cell cytoplasm	300 mmol dm^{-3}

Which of the following statements is/are correct?

- 1** A mammalian heart cell needs energy from respiration in order to obtain more potassium ions from blood plasma.
 - 2** If a yeast cell is placed in sea water then it will lose potassium ions by osmosis.
 - 3** There is no concentration gradient for potassium ions between a bacterial cell and sea water.
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3